Chenxi Han

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Research Profile

My research focuses on **reinforcement learning**, **imitation learning**, and **manipulation models** for humanoid robots. I aim to enable robots to perform tasks more accurately and quickly in real-world settings while continually learning and improving. I co-founded **ZERITH**, where I work on humanoid motion control and robot learning.

Education

Tsinghua University, Shenzhen International Graduate School Sep. 2024 – Present
Ph.D. Student in Artificial Intelligence | Advisor: Prof. Houde Liu

Harbin Institute of Technology, Shenzhen, China Sep. 2020 – Jun. 2024
B.Eng. in Mechanical Design, Manufacturing, and Automation | Rank: 1/68; Advisors: Prof. Wenfu Xu and Prof. Jiaole Wang

Research and Engineering Experience

Tsinghua AI & Robot Lab, Tsinghua University Sep. 2024 – Present
Ph.D. Researcher | Humanoid Reference Motion Generation, Whole-Body Control, and Sim-to-Real Transfer
Focus: human-like reference motion generation, whole-body control, and Sim-to-Real transfer.
Outputs: **PRIOR** (arXiv 2026), **PMG** (ICRA 2026).

ZERITH Robotics Co., Ltd. Apr. 2024 – Present
Co-Founder and Head of Motion Control and Post-Training | Teleoperation, RL Training, and Manipulation Models
Focus: led **teleoperation systems**, **RL training**, and **manipulation model post-training** for early-stage humanoid algorithm development and deployment.

Shanghai University Oct. 2023 – Apr. 2024
Visiting Researcher | Humanoid Robot Structure Design, Control, and System Validation
Focus: led structure design, control-method development, and system validation for an in-house rigid-flex hybrid humanoid robot.
Output: **SERL** (IROS 2024).

Selected Publications and Manuscripts

[1] **PRIOR: Perceptive Learning for Humanoid Locomotion with Reference Gait Priors** 2026
Chenxi Han[*], Shilu He[*], Yi Cheng, Linqi Ye, Houde Liu[†]
arXiv preprint. **Parameterized gait priors and terrain perception** for natural, terrain-adaptive humanoid locomotion.

[2] **PMG: Parameterized Motion Generator for Human-Like Locomotion Control** 2026
Chenxi Han[*], Yuheng Min[*], Zihao Huang, Ao Hong, Hang Liu, Yi Cheng, Houde Liu[†]
IEEE International Conference on Robotics and Automation (ICRA). **Parameterized motion generator** using compact parameterized motion data to synthesize reference trajectories for imitation- and reinforcement-learning-based humanoid locomotion.

[3] **Structural Optimization of Lightweight Bipedal Robot via SERL** 2024
Yi Cheng[*], **Chenxi Han**[*], Yuheng Min, Linqi Ye, Houde Liu[†], Hang Liu
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). **RL and evolutionary search** for lightweight bipedal robot design and performance improvement.

[4] **Safe Expeditious Whole-Body Control of Mobile Manipulators for Collision Avoidance** 2025
Bingjie Chen, Yancong Wei, Rihao Liu, **Chenxi Han**, Houde Liu[†], Chongkun Xia, Liang Han, Bin Liang
Biomimetic Intelligence and Robotics. **CBF-QP whole-body controller** for real-time safe collision avoidance in dynamic scenes.

[5] **Visual Servoing-Based Pneumatic Hair Transplantation Mechanism for Robotic FUE Surgery** 2023
Fulin Jia, Shenghao Yang, **Chenxi Han**, Junye Li, Xuanru Han, Chao Zhang, Jiaole Wang[†]
Biomimetic Intelligence and Robotics. **Pneumatic mechanism and visual feedback** for high-precision automated FUE surgery.

[*] Equal contribution. [†] Corresponding author.

Honors and Academic Service

Outstanding Graduate, HIT Shenzhen; **Dean's Scholarship** (**Top 1%**) 2024
Undergraduate National Scholarship, Ministry of Education of China (**Top 0.2%**) 2021
Global 4th Place, IEEE ICRA Quadruped Robot Challenge 2024
National First Prize, "Higher Education Cup" Advanced Mapping Competition 2022
Software Copyright Registration, "Optimization-Based Humanoid Robot Dataset Mapping Software" (**First Developer**) 2025
Reviewer, IEEE ICRA; IEEE Robotics and Automation Letters (RA-L) 2025–2026

Skills

Core Tools: **Python**, **PyTorch**, **Isaac Lab**, **Isaac Sim**, **Isaac Gym**, **MuJoCo**, Pinocchio, Pyroki, SolidWorks, AutoCAD.